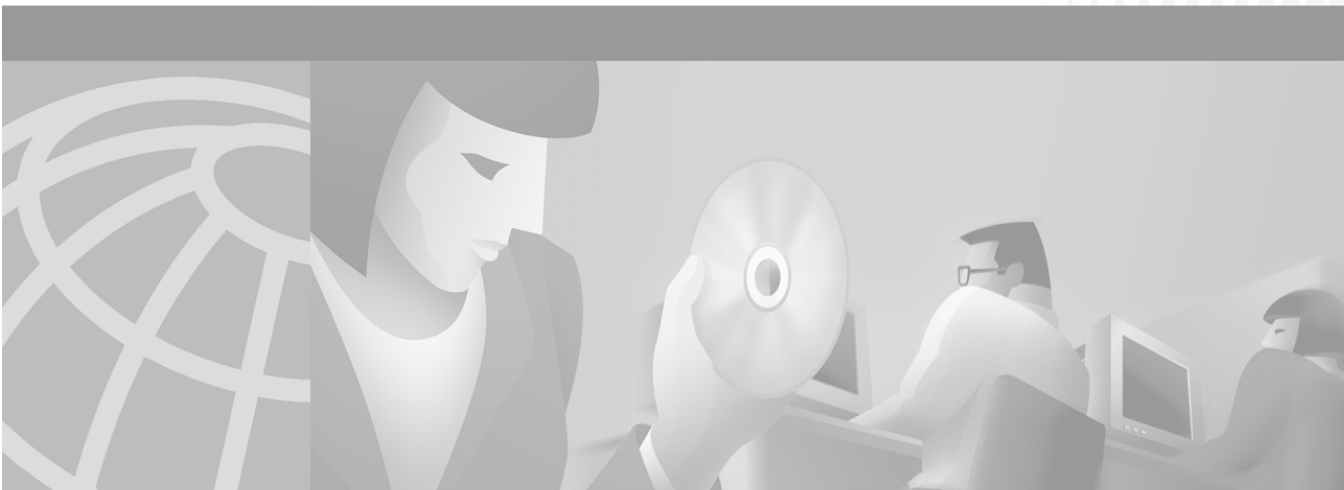




Cisco VPN Client User Guide for Linux, Solaris, and Mac OS X

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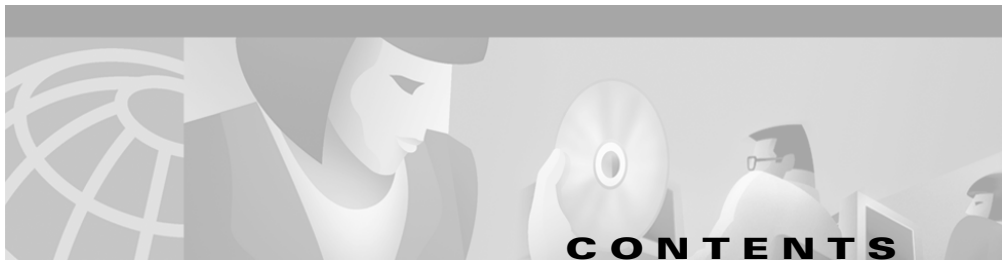
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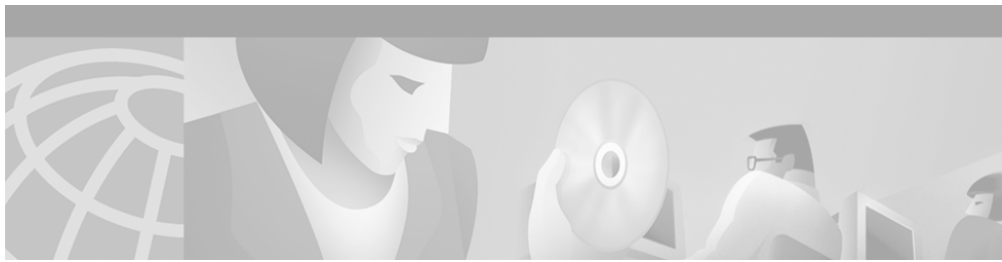
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About This Guide

This guide provides users and administrators with information about the Cisco VPN Client software for the following operating systems:

- Linux for Intel
- Solaris UltraSPARC
- Mac OS X

Contents

This guide contains the following chapters:

- Chapter 1, “Introduction to the VPN Client.” This chapter provides a brief introduction to the VPN client software.
- Chapter 2, “Installing the VPN Client.” This chapter describes how to install the VPN client software on your workstation.
- Chapter 3, “User Profiles.” This chapter describes how to set up user profiles.
- Chapter 4, “Using the Command Line Interface.” This chapter describes the command line interface and lists the commands and their descriptions.
- Chapter 5, “Managing Digital Certificates.” This chapter describes how to manage your digital certificate stores.

- Chapter 6, “Preconfiguring the VPN Client for Remote Users.” This chapter describes how administrators can preconfigure the VPN client for remote users.
- Index

Related Documentation

The following is a list of user guides and other documentation related to the VPN client for Linux.

- *Cisco VPN Client Administration Guide*
- *Cisco VPN 3000 Concentrator Series Getting Started*
- *Cisco VPN 3000 Concentrator Series User Guide*
- *Cisco VPN 5000 Concentrator Software Configuration Guide*
- *Cisco VPN 5000 Concentrator Series Command Reference Guide*

Documentation Conventions

The following typographic conventions are used in this guide.

Data Formats

When you configure the VPN client, enter data in these formats unless the instructions indicate otherwise.

- IP Address—Use standard 4-byte dotted decimal notation (for example, 192.168.12.34). You can omit leading zeros in a byte position.
- Hostnames—Use legitimate network host or end-system name notation (for example, VPN01). Spaces are not allowed. A hostname must uniquely identify a specific system on a network. A hostname can be up to 255 characters in length.

- User names and Passwords—Text strings for user names and passwords use alphanumeric characters in both upper- and lower-case. Most text strings are case sensitive. For example, `simon` and `Simon` would represent two different user names. The maximum length of user names and passwords is generally 32 characters, unless specified otherwise.

Obtaining Documentation

The following sections explain how to obtain documentation from Cisco Systems.

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You can access the most current Cisco documentation on the World Wide Web at the following URL:

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The Cisco TAC is available to all customers who need technical assistance with a Cisco product, technology, or solution. Two types of support are available through the Cisco TAC: the Cisco TAC Web Site and the Cisco TAC Escalation Center.

Inquiries to Cisco TAC are categorized according to the urgency of the issue:

- Priority level 4 (P4)—You need information or assistance concerning Cisco product capabilities, product installation, or basic product configuration.
- Priority level 3 (P3)—Your network performance is degraded. Network functionality is noticeably impaired, but most business operations continue.
- Priority level 2 (P2)—Your production network is severely degraded, affecting significant aspects of business operations. No workaround is available.
- Priority level 1 (P1)—Your production network is down, and a critical impact to business operations will occur if service is not restored quickly. No workaround is available.

Which Cisco TAC resource you choose is based on the priority of the problem and the conditions of service contracts, when applicable.

Cisco TAC Web Site

The Cisco TAC Web Site allows you to resolve P3 and P4 issues yourself, saving both cost and time. The site provides around-the-clock access to online tools, knowledge bases, and software. To access the Cisco TAC Web Site, go to the following URL:

<http://www.cisco.com/tac>

All customers, partners, and resellers who have a valid Cisco services contract have complete access to the technical support resources on the Cisco TAC Web Site. The Cisco TAC Web Site requires a Cisco.com login ID and password. If you have a valid service contract but do not have a login ID or password, go to the following URL to register:

<http://www.cisco.com/register/>

If you cannot resolve your technical issues by using the Cisco TAC Web Site, and you are a Cisco.com registered user, you can open a case online by using the TAC Case Open tool at the following URL:

<http://www.cisco.com/tac/caseopen>

If you have Internet access, it is recommended that you open P3 and P4 cases through the Cisco TAC Web Site.

Cisco TAC Escalation Center

The Cisco TAC Escalation Center addresses issues that are classified as priority level 1 or priority level 2; these classifications are assigned when severe network degradation significantly impacts business operations. When you contact the TAC Escalation Center with a P1 or P2 problem, a Cisco TAC engineer will automatically open a case.

To obtain a directory of toll-free Cisco TAC telephone numbers for your country, go to the following URL:

<http://www.cisco.com/warp/public/687/Directory/DirTAC.shtml>

Before calling, please check with your network operations center to determine the level of Cisco support services to which your company is entitled; for example, SMARTnet, SMARTnet Onsite, or Network Supported Accounts (NSA). In addition, please have available your service agreement number and your product serial number.



Introduction to the VPN Client

The Cisco VPN Client connects a remote user to a corporate network.

The user connects to a local Internet service provider (ISP), then to the VPN device Internet IP address. The VPN client encrypts the data and encapsulates it into a routable IPSec packet, creating a secure tunnel between the remote user and the corporate network.

The corporate server authenticates the user, decrypts and authenticates the IPSec packet, and translates the source address in the packets to an address recognized on the corporate network. This address is used for all traffic sent from the corporate network to the remote user for the duration of the connection.

Features

The VPN client distinguishes between tunneled and nontunneled traffic and, depending on your server configuration, allows simultaneous access to the corporate network and to Internet resources.

The VPN client communicates over async serial PPP links and Internet-attached Ethernet connections.

Table 1-1 lists VPN client features.

Table 1-1 VPN Client Features

Feature	Description
Operating systems	- Red Hat Version 6.2 Linux (Intel), or compatible distribution, using kernel Version 2.2.12 or later Note The VPN client for Linux does not support kernel Version 2.5. - Solaris UltraSPARC running a 32-bit kernel OS Version 2.6 or later - Mac OS X Version 10.1.0 or later
Connection types	- async serial PPP - Ethernet
Protocol	IP
Tunnel protocol	IPSec
User authentication	- RADIUS - RSA SecurID - NT Domain - VPN server internal user list - PKI digital certificates



Installing the VPN Client

This chapter describes how to install the VPN client software on your workstation. You should be familiar with software installation on UNIX or Macintosh computers before you perform this procedure.

The VPN client consists of:

- A driver, which is a loadable module.
- A set of commands accessible through your shell, which is used to access the applications.

The commands and some parts of the driver are distributed in binary form only.

Contents

This chapter contains the following sections:

- Uninstalling an Old Client, page 2-2
- System Requirements, page 2-3
- Unpacking the VPN Client Files, page 2-5
- Installing the Software, page 2-6

Uninstalling an Old Client

This section describes how to uninstall the VPN client.

- You *must* uninstall an old VPN client for Solaris before you install a new VPN client.
- You are *not required* to uninstall an old VPN client for Linux or for Mac OS X before you install a new VPN client.

Uninstalling a VPN Client for Solaris

If a VPN client for Solaris was previously installed, you must remove the old VPN client before you install a new one.

To uninstall a package, use the **pkgrm** command. For example:

```
pkgrm vpnclient
```

Uninstalling a VPN Client for Linux or Mac OS X

To uninstall the VPN client for Linux or Mac OS X:

- a. Locate the script `vpn_uninstall`.
This file must be run as root.
- b. You are prompted to remove all profiles and certificates.
 - If you answer yes, all binaries, startup scripts, certificates, profiles, and any directories that were created during the installation process are removed.
 - If you answer no, all binaries and startup scripts are removed, but certificates, profiles, and the `vpnclient.ini` file remain.

System Requirements

This section describes system requirements for the VPN client for each operating system.

Linux System Requirements

The VPN client for Linux supports Red Hat Version 6.2 Linux (Intel), or compatible libraries with glibc Version 2.1.1-6 or later, using kernel Versions 2.2.12 or later.

**Note**

The VPN client for Linux does not support kernel Version 2.5.

Firewall Issues

If you are running a Linux firewall (for example, ipchains or iptables), be sure that the following types of traffic are allowed to pass through:

- UDP port 500
- UDP port 10000 (or any other port number being used for IPSec/UDP)
- IP protocol 50 (ESP)
- TCP port configured for IPSec/TCP

Troubleshooting Tip

The following two lines might be added by default with your Linux installation in the /etc/sysconfig/ipchains directory. For Redhat, this might be written to the /etc/sysconfig/ipchains directory. These two commands might prevent UDP traffic from passing through.

```
-A input -p udp -s 0/0 -d 0/0 0:1023 -j REJECT  
-A input -p udp -s 0/0 -d 0/0 2049 -j REJECT
```

If you have problems with UDP traffic, first delete the above two lines, then enter the following two commands:

```
/etc/init.d/ipchains stop
/etc/init.d/ipchains start
```


Note

Ipchains might be replaced by iptables or it might be located in a different directory on your Linux distribution.

Solaris System Requirements

The VPN client for Solaris runs on any ultraSPARC computer running a 32-bit Solaris kernel OS Version 2.6 or later.

Using the 32-Bit Kernel

Some Solaris machines run a 64-bit kernel by default. To use the VPN client, run the 32-bit version of the kernel.

There are several ways to run in 32-bit mode.

- Specify the kernel/unix as the boot file. Enter the following command:

```
ok boot kernel/unix
```

This command immediately reboots the system in 32-bit mode. 32-bit mode is only valid for this boot. When you reboot again, the system switches back to its default mode.

- Switch to 32-bit mode permanently. Enter the following command:

```
eeeprom boot-file=/platform/sun4u/kernel/unix
```

You must reboot after you issue this command.

- Switch back to 32-bit mode permanently. Enter the following command:

```
eeeprom boot-file=/platform/sun4u/kernel/sparcv9/unix
```

You must reboot after you issue this command.

To confirm that your system is running in 32-bit mode:

- a. Issue the following command:

```
isainfo -kv
```

- b. When the Solaris system boots up, a message in the **dmesg** event log similar to the following appears:

```
Oct 29 11:09:54 sol-2062 cipsec: [ID 952494 kern.notice] Cisco  
Unity IPSec Module Load OK
```

If you do not receive this message, the IPSec module did not load properly and you need to switch to the 32-bit kernel.

Mac OS X System Requirements

The VPN client for Mac OS X runs on any Macintosh computer running OS X Version 10.1.0 or later.



Note

Classic Mac applications do not make use of the VPN tunnel.

Unpacking the VPN Client Files

The VPN client is shipped as a compressed tar file.

For Solaris, there are two available VPN client files. Make sure that you have the correct installation file for your operating system.

- The installation file for Solaris 5.6 and Solaris 7 is named:

```
vpnclient-solaris5.6-3.5.xxx-K9.tar.Z
```

- The installation file for Solaris 8 is named:

```
vpnclient-solaris5.8-3.5.xxx-K9.tar.Z
```

To unpack the files

-
- Step 1** Download the packed files, either from your internal network or the Cisco website, to a directory of your choice.
 - Step 2** Copy the VPN client file to a selected directory.
 - Step 3** Unpack the file using the **zcat** and **tar** commands.

For example, the command for Linux is:

```
zcat vpnclient-linux-3.5.xxx-K9.tar.gz | tar xvf -
```

The command for SPARC Solaris is:

```
zcat vpnclient-solaris5.8-3.5.xxx-K9.tar.Z | tar xvf -
```

The command for Mac OS X is:

```
zcat vpnclient-macosx-3.5.xxx-k9.tar.gz | tar xvf -
```

This command creates the vpnclient directory in the current directory.

Installing the Software

The following sections describe the installation procedure for the VPN client for each operating system.



Note

You cannot have both a VPN 5000 client and a Unified VPN client installed on your workstation. You must uninstall one before you use the other. Refer to the “Uninstalling an Old Client” section on page 2-2 for more information.

Installing the VPN Client for Linux

Before you install a new version of the VPN client, or before you re-install your current version, you must use the **stop** command to disable VPN service.

If you are upgrading from the VPN 5000 client to the VPN client, use the following **stop** command:

```
/etc/rc.d/init.d/vpn stop
```

If you are upgrading from the VPN 3000 client to the VPN client, use the following **stop** command:

```
/etc/rc.d/init.d/vpnclient_init stop
```

To install the VPN client for Linux

-
- Step 1** Obtain superuser privileges to run the install script.
- Step 2** Enter the following commands:
- ```
cd vpnclient
./vpn_install
```
- Step 3** At the prompt, choose a directory in which to install the VPN client. Use the default directory (by pressing **Enter**), or choose a directory in your user's path.
- Step 4** Enable the VPN service by using one of the following methods:
- Reboot your computer.
  - Enable the service without rebooting. Enter the following command:
- ```
/etc/rc.d/init.d/vpnclient_init start
```
-

VPN Client for Linux Install Script Notes

During the installation process:

1. The module is compiled, linked, and copied to either the directory `/lib/modules/preferred/CiscoVPN`, if it exists, or to `/lib/modules/system/CiscoVPN`, where *system* is the kernel version.
2. The application binaries are copied to the specified destination directory.
3. The startup file `/etc/rc.d/init.d/vpnclient_init` is created to enable and disable the VPN service.

4. The links `/etc/rc3.d/s85vpncclient` and `/etc/rc5.d/s85vpncclient` are added to run level 3 and level 5 if startup at boot time is requested.

These links allow the tunnel server to start at boot time and run in levels 3 and 5.

Installing the VPN Client for Solaris

Before you install a new version of the VPN client, or before you re-install your current version, you must uninstall the old VPN client. See the “Uninstalling an Old Client” section on page 2-2 for more information.

To install the VPN client for Solaris

-
- | | |
|---------------|---|
| Step 1 | Obtain superuser privileges to run the install script. |
| Step 2 | Enter the following command:

<code>pkgadd -d . vpncclient</code> |
| Step 3 | At the prompt, choose a directory in which to install the VPN client applications.
Use the default directory (by pressing Enter), or choose a directory in your user's path. |
| Step 4 | Respond Yes to any other prompts to complete the installation. |
| Step 5 | Reboot your computer. |
-

VPN Client for Solaris Install Script Notes

During the installation process:

1. The following line is added to the `/etc/iu.ap` file to enable the autopush facility at startup:

`hme -1 0 cipsec`
2. The VPN module is copied to the `/kernel/strmod` directory, which is in the system's module search path.

The **pkginfo** command provides information about the installed packages. For more information on other package-related commands, enter:

```
man pkgadd
```

Installing the VPN Client for Mac OS X

**Note**

You must have root privileges to install the VPN client for Mac OS X.

To install the VPN client for Mac OS X

Step 1 Activate the root account.

The root account is disabled by default. Open the application NetInfo Manager in the Utilities folder, which is in the Applications folder. Click the button with the lock and enter your password. In the menu choose **Domain > Security > Authenticate** and then **Domain > Security > Enable Root User**. You are prompted for a password.

Step 2 Obtain superuser privileges to run the install script.

Step 3 Enter the following commands:

```
cd vpnclient
./vpn_install
```

Step 4 At the prompt, choose a directory in which to install the VPN client.

Use the default directory (by pressing Enter), or choose a directory in your user's path.

Step 5 Respond to the question about automatically loading the VPN NKE at boot time.

- If you answer **Yes**, use the following commands to control the NKE:

```
/System/Library/StartupItems/CiscoVPN/CiscoVPN start
/System/Library/StartupItems/CiscoVPN/CiscoVPN stop
/System/Library/StartupItems/CiscoVPN/CiscoVPN restart
```

- If you answer **No**, use the following commands to control the NKE:

```
kmodload  
/System/Library/Extensions/CiscoVPN.kext/Contents/MacOS/CiscoVPN  
  
kmodunload com.cisco.nke.ipsec
```

VPN Client for Mac OS X Install Script Notes

During the installation process:

1. The application binaries are copied to the specified destination directory.
2. Use the following commands to start, stop, and restart VPN service:
 - /System/Library/StartupItems/CiscoVPN/CiscoVPN start
 - /System/Library/StartupItems/CiscoVPN/CiscoVPN stop
 - /System/Library/StartupItems/CiscoVPN/CiscoVPN restart



User Profiles

This chapter describes how to create a VPN client user profile. A user profile is a list of configuration keywords that determine the connection entries for a remote user.

There are two ways to create a user profile:

- Use a text editor to modify the sample profile that comes with the VPN client installer and rename it.
- Create a unique user profile using a text editor.

User profiles have a .pcf file extension and reside in the default location `/etc/CiscoSystemsVPNClient/Profiles/` directory.

There is only one user profile per connection.



Tip

User profiles for the VPN client are interchangeable between platforms.

Contents

This chapter includes the following sections:

- Sample Profile, page 3-2
- Modifying the Sample Profile, page 3-2
- User Profile Keywords, page 3-3

Sample Profile

The VPN client software is shipped with a sample user profile. The file is named `sample.pcf`.

The following is an example of a sample user profile that might be shipped with your installer.

```
[main]
Description=sample user profile
Host=10.7.44.1
AuthType=1
GroupName=monkeys
EnableISPConnect=0
ISPConnectType=0
ISPConnect=
ISPCommand=
Username=gawf
SaveUserPassword=0
EnableBackup=0
BackupServer=
EnableNat=0
CertStore=0
CertName=
CertPath=
CertSubjectName=
CertSerialHash=00000000000000000000000000000000
DHGroup=2
ForceKeepAlives=0
```

Modifying the Sample Profile

To modify the sample profile

-
- Step 1** Using a text editor, open the sample user profile.
 - Step 2** Modify the keywords you want to change.
See your administrator for IP addresses, user name, and any security information.
 - Step 3** Save your new profile with a unique name in the
`/etc/CiscoSystemsVPNClient/Profiles/` directory.

When you use the **vpnclient connect** command to establish a connection, use your new profilename.

User Profile Keywords

You can create your own user profile from scratch using any text editing program.

At a minimum, you need the following keywords listed in your profile:

- [main]
- Host
- AuthType
- GroupName
- Username

Save your new profile in the `/etc/CiscoSystemsVPNClient/Profiles/` directory. See your administrator for IP addresses, user name, and any security information.

Table 3-1 describes the keywords that can be in a user profile. User profile keywords are not case sensitive unless indicated in the description.

Table 3-1 User Profile Keywords

Keywords	Description
[main]	This keyword is required and is used to identify the main section. Enter exactly as shown as the first entry in the user profile.
Description = <i>String</i>	This optional command describes this user profile. The maximum length is 246 alphanumeric characters.
Host = <i>IP_Address or hostname</i>	The hostname or IP address of the VPN device you want to connect with. The maximum length of the hostname is 255 alphanumeric characters.

Table 3-1 *User Profile Keywords (continued)*

Keywords	Description
AuthType = { 1 3 }	The authentication type that this user is using. • 1 is preshared keys. • 3 is a digital certificate using an RSA signature. If you select AuthType 1, you must also configure the GroupName and GroupPwd.
GroupName = <i>String</i>	The name of the IPsec group configured on the VPN device that contains this user. The maximum length is 32 alphanumeric characters. This keyword is case sensitive.
GroupPwd = <i>String</i>	The password for the IPsec group that contains this user. The minimum length is 4 alphanumeric characters. The maximum is 32. This keyword is case sensitive and entered in clear text.
encGroupPwd = <i>String</i>	This keyword displays the group password in the user profile in its encrypted form. It is binary data represented as alphanumeric text.
Username = <i>String</i>	The name that identifies a user as a valid member of the IPsec group specified in GroupName. The VPN client prompts the user for this value during user authentication. The maximum length is 32 alphanumeric characters. This keyword is case sensitive and entered in clear text.

Table 3-1 User Profile Keywords (continued)

Keywords	Description
UserPassword = <i>String</i>	This password is used during extended authentication. If SaveUserPassword is enabled, the first time the VPN client reads this password, it is saved in the user profile as encUserPassword, and the clear text version is deleted. If SaveUserPassword is disabled, the VPN client deletes the clear text version of the user password in the user profile but it does not create an encrypted version.
encUserPassword = <i>String</i>	This keyword displays the user password in the user profile in its encrypted form. It is binary data represented as alphanumeric text.
SaveUserPassword = { 0 1 }	Determines if the user password or its encrypted form are valid in the user profile. • 0, the default, displays the user password in clear text in the user profile and is saved locally. • 1 displays the user password in the user profile in its encrypted version, and the password is not saved locally. This value is set in the VPN device, not in the VPN client.
EnableBackup = { 0 1 }	Specifies to use a backup server if the primary server is not available. • 0, the default, disables the backup server. • 1 enables the backup server. You must also specify a BackupServer.

Table 3-1 User Profile Keywords (continued)

Keywords	Description
BackupServer = <i>IP_Address or hostname</i>	List of IP addresses or hostnames of backup servers. Separate multiple entries by commas. The maximum length of hostname is 255 alphanumeric characters.
EnableLocalLAN = { 0 1 }	This keyword allows you to configure access to your local LAN. • 0, the default, disables local LAN access. • 1 enables local LAN access.
EnableNAT = { 0 1 }	This keyword specifies whether or not to enable secure transmission between a VPN client and a VPN device through a router serving as a firewall, which might also be using the NAT protocol. • 0, the default, disables IPsec through NAT mode. • 1 enables IPsec through NAT mode.
TunnelingMode = { 0 1 }	This keyword allows you to select which form of NAT transversal is used. • 0, the default, specifies IPsec over UDP for NAT transparency. • 1 specifies IPsec over TCP for NAT transparency. You must also have IPsec through NAT enabled.
TCTunnelingPort = { 0 65535 }	This keyword sets which TCP port to use for the cTCP protocol. The default is 10000. You must also have IPsec through NAT enabled and the Tunneling Mode set for IPsec over TCP.

Table 3-1 *User Profile Keywords (continued)*

Keywords	Description
ForceKeepAlives = { 0 1 }	This keyword allows the VPN client to keep sending IKE and ESP keepalives for a connection at approximately 20-second intervals so that the port on an ESP-aware NAT/Firewall does not close. • 0, the default, disables keepalives. • 1 enables keepalives.
PeerTimeout = <i>Number</i>	The number of seconds to wait before terminating a connection when the VPN device on the other end of the tunnel is not responding. The range is 30 to 480 seconds. The default is 90.
CertStore = { 0 1 }	Identifies the type of store containing the configured certificate. • 0 = default, none. • 1 = Cisco.
CertName = <i>String</i>	Identifies the certificate used to connect to the VPN device. The maximum length is 129 alphanumeric characters.
CertPath = <i>String</i>	The path name of the directory containing the certificate file. The maximum length is 259 alphanumeric characters.
CertSubjectName = <i>String</i>	The qualified distinguished name (DN) of the certificate's owner. You can either <i>not</i> include this keyword in the user profile, or leave this entry blank.

Table 3-1 User Profile Keywords (continued)

Keywords	Description
CertSerialHash = <i>String</i>	A hash of the certificate’s complete contents, which provides a means of validating the authenticity of the certificate. You can either <i>not</i> include this keyword in the user profile, or leave this entry blank.
DHGroup = { 1 2 }	Allows a network administrator to override the configured group value used to generate Diffie-Hellman key pairs on a VPN device. • 1 = modp group 1 • 2 = modp group 2 The default is 2. The VPN Concentrator configuration for IKE Proposal must match the DHGroup in the VPN client. If the AuthType is set to 3 (digital certificate), this keyword has no effect on the VPN client.



Using the Command Line Interface

This chapter explains how to use the VPN client's command line interface (CLI) to connect to a Cisco VPN device, generate statistical reports, and disconnect from the device. You can create your own script files that use the CLI commands to perform routine tasks, such as connect to a corporate server, run reports, and then disconnect from the server.

Contents

This chapter contains the following sections:

- Displaying a List of VPN Client Commands, page 4-1
- Establishing a Connection, page 4-2
- Logging Files, page 4-4
- Disconnecting the VPN Client, page 4-5
- Displaying VPN Client Statistics, page 4-5

Displaying a List of VPN Client Commands

To display a list of available VPN client commands, go to the directory that contains the VPN client software and enter the **vpnclient** command at the command line prompt.

The following example shows the command and the information that is displayed.

```
%vpnclient
Cisco Systems VPN Client Version 3.0.7
Copyright (C) 1998-2001 Cisco Systems, Inc. All Rights Reserved.
Client Type(s): Linux
Running on: Linux 2.2.14-5.0 #1 Tue Mar 7 21:07:39 EST 2000 i686
```

Usage:

```
vpnclient connect profilename [nocertpwd] [eraseuserpwd]
vpnclient disconnect
vpnclient stat [reset] [traffic] [tunnel] [route] [repeat]
```

Establishing a Connection

This section describes how to establish a VPN connection, parameters you might need to enter, and how to manipulate the VPN Client window.



Note

If you are connecting to a VPN device using Telnet or SSH, check to see if the device allows split tunneling. If it does not, you lose connectivity to your VPN device after making a VPN connection.

To establish a connection, enter the following command:

```
vpnclient connect profilename [nocertpwd] [eraseuserpwd]
```

Profilename is the name of the user profile configured for this user (.pcf file). This parameter is required. Enter your profilename without the .pcf file extension. If your profilename contains spaces, enclose it in double quotation marks on the command line.

If your user profile is configured with the **SaveUserPassword** keyword set to the default, the password is saved locally. The **eraseuserpwd** keyword erases the user password that is saved on the VPN client workstation, forcing the VPN client to prompt you for a password. The **eraseuserpwd** keyword is an optional parameter that returns the VPN client to a state that requires you to enter a password each time you try to establish a connection.

The **nocertpwd** is a keyword that suppresses the prompt for a certificate password.

For more information on `profilename` keywords, see the “User Profile Keywords” section on page 3-3.

Depending on the parameters that have been configured in your user profile, you are prompted for the following passwords:

- Group password
- User name
- User password

If your VPN client has been configured to use SecurID or RADIUS authentication, you are prompted for those passwords.

See your administrator for any security information.

When the connection is established, the VPN Client window stays in the foreground to allow the VPN client to be reauthenticated during a rekey by the VPN device. To send the VPN Client window to the background, press **Ctrl-Z** followed by the **bg** command at the command line prompt.

Rekeying Issues

This section describes rekeying issues between the VPN client and the concentrator.

If the VPN device you are connecting to is configured to support rekeying and you send the VPN Client window to the background, the tunnel disconnects when the first rekey occurs.

The VPN client responds to rekey triggers based on *time* not *data*. If you want VPN client connections rekeyed, you must configure the concentrator so that the IKE proposal is set to rekey every 1800 seconds and IPSec parameters are set to rekey every 600 seconds.

DNS Servers

You can configure the concentrator to send the IP addresses of DNS servers to the VPN client to use during tunnel sessions.

If the client receives the DNS server settings, it copies the file `/etc/resolv.conf` to a backup file `/etc/resolv.conf.vpnbackup`. When the tunnel closes, the original contents of `/etc/resolv.conf` are restored.

Logging Files

This section provides information on the logging files, including how to capture and view logging information.

To enable logging, you must edit the `/etc/CiscoSystemsVPNClient/vpnclient.ini` file to include the following:

```
[main]
EnableLog=1
[LOG.IKE]
LogLevel=3
[LOG.CM]
LogLevel=3
[LOG.PPP]
LogLevel=3
[LOG.CVPND]
LogLevel=3
[LOG.CERT]
LogLevel=3
[LOG.IPSEC]
LogLevel=3
```

For more information on logging, see the “Global Profile Configuration Parameters” section on page 6-3.

To view logging information, enter the following command:

```
/usr/local/bin/ipseclog /directory/clientlog.txt
```



Note

If you did not use the default directory `/usr/local/bin` during installation, logging commands must be entered using your chosen path.

To view logging information in real time, enter the following command after you start the `ipseclog`:

```
tail -f /directory/clientlog.txt
```

The `ipseclog` does not automatically go to the background. To send the `ipseclog` to the background, press **Ctrl-Z** followed by **bg** on the command line, or enter the ampersand symbol at the end of the **view** command, as shown in the following example:

```
/usr/local/bin/ipseclog /directory/clientlog.txt &
```

If the ipseclog is in the background, you must send it to the foreground before you end the VPN client application. To send the ipseclog to the foreground, enter **fg** on the command line.

Client Auto Update Messages

When the VPN client receives an auto-update notification from the VPN remote access device, it logs the notification, but takes no further action. The message logged upon receipt of an auto-update message has the following format:

```
688 10:14:23.811 08/08/2001 Sev=Info/4 IKE/0x4300005B^M
CLIENT UPDATE NOTIFICATION: Client Type="linux" Revisions="3.0.5"
URL="tftp://x:/y/z"
```

Disconnecting the VPN Client

This section describes methods for disconnecting the VPN client.

To disconnect from your session, use one of the following methods:

- Enter the following command:

```
vpnclient disconnect
```

The following example shows the command that disconnects you from your secure connection and the prompt that appears when you are not connected.

```
vpnclient disconnect
Disconnecting the IPSEC link.
Your IPsec link is not connected.
```

- Press **Ctrl-C** while you are in the VPN Client window.

Displaying VPN Client Statistics

This section describes the VPN client statistics command and the optional parameters to the command.

To generate status information about your connection, enter the following command:

```
vpnclient stat [reset] [traffic] [tunnel] [route] [repeat]
```

If you enter this command without any of the optional parameters, the **vpnclient stat** command displays all status information. The optional parameters are described in Table 4-1.

Table 4-1 *Optional Parameters to the VPN Client Stat Command*

Parameter	Description
reset	Restarts all connection counts from zero.
traffic	Displays a summary of bytes in and out, packets encrypted and decrypted, and packets discarded.
tunnel	Displays IPSec tunneling information.
route	Displays configured routes.
repeat	Provides a continuous display, refreshing it every few seconds. To end the display, press Ctrl-C .

Examples

This section shows examples of output from the different options for the **vpnclient stat** command.

No Options

The following is a sample output from the **vpnclient stat** command with no options.

```
vpnclient stat

IPSec tunnel information.
Client address: 209.154.64.50
Server address: 10.10.32.32
Encryption: 168-bit 3-DES
Authentication: HMAC-MD5
IP Compression: None
NAT passthrough is active on port 5000
```

```
VPN traffic summary.  
Time connected: 0 day<s>, 00:18.32  
Bytes out: 3420  
Bytes in: 3538  
Packets encrypted: 23  
Packets decrypted: 57  
Packets bypassed: 102  
Packets discarded: 988
```

```
Configured routes  
Secured   Network Destination   Netmask           Bytes  
*         10.10.32.32             255.255.255.255   7638  
*         0.0.0.0                 0.0.0.0           1899
```

Reset Option

To reset all connection counters, use the **vpnclient stat reset** command.

```
vpnclient stat reset  
Tunnel statistics have been reset.
```

Traffic Option

The following is a sample output from the **vpnclient stat** command with the **traffic** option.

```
vpnclient stat traffic  
  
VPN traffic summary  
Time connected: 0 day<s>, 00:30:04  
Bytes out: 5460  
Bytes in: 6090  
Packets encrypted: 39  
Packets decrypted: 91  
Packets bypassed: 159  
Packets discarded: 1608
```

Tunnel Option

The following is a sample output from the **vpnclient stat** command with the tunnel option. The **vpnclient stat tunnel** command shows only tunneling information.

```
vpnclient stat tunnel

IPSec tunnel information.
Client address: 220.111.22.30
Server address: 10.10.10.1
Encryption: 168-bit 3-DES
Authentication: HMAC-MD5
IP Compression: None
NAT passthrough is active on port 5000
```

Route Option

The following is a sample output from the **vpnclient stat** command with the route option.

```
vpnclient stat route

Configured routes
Secured  Network Destination      Netmask          Bytes
*        10.10.02.02             255.255.255.255   17638
*        0.0.0.0                0.0.0.0           18998
```



Managing Digital Certificates

This chapter describes how to manage digital certificates in your certificate store for the Cisco VPN client using the command line interface. Your certificate store is the location in your local file system for storing digital certificates. The store for the VPN client is the Cisco store.

Contents

This chapter includes the following sections:

- User Profile Keywords, page 5-2
- Command Line Interface, page 5-2
- Certificate Contents, page 5-3
- Password Protection on Certificates, page 5-5
- Certificate Management Operations, page 5-5
- Certificate Tags, page 5-8
- Enrolling Certificates, page 5-9

User Profile Keywords

To use certificates for authentication, you must correctly set all keywords that apply to certificates in your user profile. Check your settings for the following keywords:

- **AuthType = 3** (certificate authentication)
- **CertStore = 1** (Cisco certificate store)
- **CertName = Common Name** (This must be the same common name that is entered for a certificate.)

See Chapter 3, “User Profiles,” for more information on setting parameters in your user profile.

Command Line Interface

Digital certificate management is implemented using the command line interface.

The command line interface for certificate management operates in two ways:

- The standard UNIX shell at which you enter all arguments for a given command on the same line.

```
cisco_cert_mgr -U -op enroll -f filename -chall challenge_phrase
```

- A prompting mode in which you enter minimum arguments for a given command and are prompted for any remaining information.

The minimum command line argument follows this basic form:

```
cisco_cert_mgr -U -op operation
cisco_cert_mgr -R -op operation
cisco_cert_mgr -E -op operation
```

Where:

- **-U** applies to the user or private certificate.

You can use the -U flag for all certificate management command operations, except enroll_resume.

- **-R** applies the root certificate or certificate authority (CA) certificate.

You can use the -R flag for list, view, verify, delete, export, import, and change password operations.

- **-E** applies to certificate enrollment.

You can only use the **-E** flag with list and delete, and you must specify it using the `enroll_resume` operation.

The operation for the specified certificate follows the **-op** argument. Valid operations for the certificate manager command are list, view, verify, delete, export, import, enroll, `enroll_file`, and `enroll_resume`. For more information on these operations, see the “Certificate Management Operations” section on page 5-5.

Certificate Contents

This section describes the type of information contained in a digital certificate.

A typical digital certificate contains the following information:

- **Common name**—The name of the owner, usually both the first and last names. This field identifies the owner within the Public Key Infrastructure (PKI organization).
- **Department**—The name of the owner’s department. This is the same as the organizational unit.
 - If you are connecting to a VPN 3000 concentrator, this field must match the **Group Name** configured for the owner in the concentrator.
 - If you are connecting to a VPN 5000 concentrator, this field must match the **VPNGroup-groupname** configured in the concentrator.
- **Company**—The company in which the owner is using the certificate. This is the same as the organization.
- **State**—The state in which the owner is using the certificate.
- **Country**—The 2-character country code in which the owner’s system is located.
- **Email**—The e-mail address of the owner of the certificate.
- **Thumbprint**—An MD5 hash of the certificate’s complete contents. This provides a means for validating the authenticity of the certificate. For example, if you contact the issuing CA, you can use this identifier to verify that this certificate is the correct one to use.
- **Key size**—The size of the signing key pair in bits.

- **Subject**—The fully qualified distinguished name (FQDN) of the certificate's owner. This field uniquely identifies the owner of the certificate in a format that can be used for LDAP and X.500 directory queries. A typical subject includes the following fields:
 - common name (**cn**)
 - organizational unit, or department (**ou**)
 - organization or company (**o**)
 - locality, city, or town (**l**)
 - state or province (**st**)
 - country (**c**)
 - e-mail address (**e**)

Other items might be included in the Subject, depending on the certificate.

- **Serial number**—A unique identifier used for tracking the validity of the certificate on the Certificate Revocation Lists (CRLs).
- **Issuer**—The fully qualified distinguished name (FQDN) of the source that provided the certificate.
- **Not before**—The beginning date that the certificate is valid.
- **Not after**—The end date beyond which the certificate is no longer valid.

The following output is an example of the type of information contained in a digital certificate:

```
Common Name: Fred Flinstone
Department: Rock yard
Company: Stone Co.
State: (null)
Country: (null)
Email: fredf@stonemail.fake
Thumb Print: 2936A0C874141273761B7F06F8152CF6
Key Size: 1024
Subject: e=fredf@stonemail.fake,cn=Fred Flinstone,ou=Rockyard,o=Stone
Co. l=Bedrock
Serial #: 7E813E99B9E0F48077BF995AA8D4ED98
Issuer: Stone Co.
Not before: Thu May 24 18:00:00 2001
Not after: Mon May 24 17:59:59 2004
```

Password Protection on Certificates

Each digital certificate is protected by a password. Many operations performed by the certificate management command require that you enter the password before the operation can take place.

The operations that require you to enter a password are:

- Delete
- Import
- Export
- Enroll

**Note**

For the enroll operation, the password to protect the digital certificate is a separate password from the optional challenge password that you enter for the server certificate.

You are prompted for any passwords that are required to complete the command. You must enter the password and verify the password again before the command can execute. If the password is not accepted, you must re-enter the command.

When you establish a VPN connection with a certificate, a certificate password is also required.

All passwords can be up to 32 alphanumeric characters in length, and are case sensitive.

Certificate Management Operations

List all certificate management operations on the command line following the minimum command line argument. Valid operation strings allow you to list, view, verify, delete, export, import, and enroll digital certificates in your store.

The following is an example of a certificate management command with the **list** operation, and a sample output.

```
cisco_cert_mgr -U -op list

cisco_cert_mgr Version 3.0.7

      Cert #           Common Name
      ----
      0             Fred Flinstone
      1              Dino
```

Table 5-1 describes the operations that can be used with the certificate management command.

Table 5-1 Certificate Management Operations

Operation	Description
list	Lists all certificates in the certificate store. Each certificate in the list is identified by a unique certificate tag (<i>Cert #</i>).
view -ct <i>Cert #</i>	Views the specified certificate. You must enter a certificate tag.
verify -ct <i>Cert #</i>	Verifies that the specified certificate is valid. You must enter a certificate tag. If the certificate is verified, the message ‘Certificate <i>Cert #</i> verified’ appears. If the certificate fails verification for any reason, the message ‘Certificate <i>Cert #</i> failed verification’ appears. Following this message a text string which describes the reason for the failure appears.
delete -ct <i>Cert #</i>	Deletes the specified certificate. You must enter a certificate tag.
export -ct <i>Cert # -f filename</i>	Exports the identified certificate from the certificate store to a specified file. You must enter a certificate tag and a filename. If either is omitted, the command line prompts you for them. You must enter the full path of the destination. If you enter only the filename, the file is placed in your working directory.

Table 5-1 *Certificate Management Operations (continued)*

Operation	Description
import -f <i>filename</i>	Imports a certificate from a specified file to the certificate store. This operation requires two different passwords: the password that protects the file (assigned by your administrator), and the password you select to protect the certificate.
enroll -cn <i>common_name</i> -ou <i>organizational_unit</i> -o <i>organization</i> -st <i>state</i> -c <i>country</i> -e <i>email</i> -ip <i>IP_Address</i> -dn <i>domain_name</i> -caurl <i>url_of_CA</i> -cadn <i>domain_name</i> [-chall <i>challenge_phrase</i>]	For user certificates only. Obtains a certificate by enrolling you with a Certificate Authority (CA) over the network. Enter each keyword individually on the command line. See the “Certificate Tags” section on page 5-8 for more information. You can obtain a challenge phrase from your administrator or from the CA.
enroll_file -cn <i>common_name</i> -ou <i>organizational_unit</i> -o <i>organization</i> -st <i>state</i> -c <i>country</i> -e <i>email</i> -ip <i>IP_Address</i> -dn <i>domain_name</i> -f <i>filename</i> -enc [base64 binary]	For user certificates only. Generates an enrollment request file that can be e-mailed to the CA or pasted into a webpage form. When the certificate is generated by the CA, you must import it using the import operation. See the “Certificate Tags” section on page 5-8 for more information.

Table 5-1 *Certificate Management Operations (continued)*

Operation	Description
enroll_resume -E -ct <i>Cert #</i>	This operation cannot be used with user or root certificates. Resumes an interrupted network enrollment. You must enter the -E argument and a certificate tag.
changepassword -ct <i>Cert #</i>	Changes a password for a specified digital certificate. You must enter a certificate tag. You must enter the current password before you select the new password and confirming it.

Certificate Tags

A certificate tag is the identifier for each unique certificate. Each certificate added to the certificate store is assigned a certificate tag. An enroll operation also generates a certificate tag, even if the enroll operation does not complete.

Some certificate management operations require that you enter a certificate tag argument before the operation can take place. Operations that require certificate tags are listed in Table 5-1. Use the **list** operation to find your certificate tag.

To enter a certificate tag argument, use the **-ct** command followed by the certificate identifier, listed as **-ct *Cert #*** next to the operation.

The following example shows the **view** command with a required certificate tag:

```
cisco_cert_mgr -U -op view -ct 0
```

Where the operation is **view**, and the certificate tag is **0**.

If you do not enter the **-ct** argument and certificate tag, the command line prompts you for them. If you enter an invalid certificate tag, the command line lists all certificates in the certificate store, and prompts you again for the certificate tag.

Enrolling Certificates

A Certificate Authority (CA) is a trusted organization that issues digital certificates to users to provide a means for verifying that a user is who they claim to be. The certificate enrollment operations allow you to obtain your certificate from a CA over the network or from an enrollment request file.

There are three types of certificate enrollment operations.

- The **enroll** operation allows you to obtain a certificate by enrolling with a CA over the network. You must enter the url of the CA, the domain name of the CA, and the common name.
- The **enroll_file** operation generates an enrollment request file that you can e-mail to a CA or post into a webpage form. You must enter a filename, a common name and the encoding type you want to use.

With the **enroll** and **enroll_file** operations, you can include additional information with associated keywords. These keywords are described in Table 5-2.

- The **enroll_resume** operation resumes an interrupted network enrollment. You must enter the **-E** argument and a certificate tag. To find your certificate tag, use the **list** operation.

Enrollment Operations

To use enrollment operations, enter the certificate manager command and the enroll operation you want to use with the associated keywords on the command line.

The following example shows the **enroll** command with the minimum required keywords for common name (**-cn**), url of the CA (**-caurl**) and domain name of the CA (**-cadn**).

```
cisco_cert_mgr -U -op enroll -cn Ren Hoek -caurl  
http://172.168.0.32/certsrv/mscep/mscep.dll -cadn nobody.fake
```

The following example shows the **enroll_file** command with the minimum required keywords for filename (**-f**), common name (**-cn**), and encoding type (**-enc**).

```
cisco_cert_mgr -U -op enroll_file -f filename -cn Ren Hoek -enc base64
```

The following example shows the `enroll_file` command with the required minimum arguments and additional keywords.

```
cisco_cert_mgr -U -op enroll_file -f filename -cn Ren Hoek -ou
Customer Service -o Stimpy, Inc, -st CO -c US -e ren@fake.fake -ip
10.10.10.10 -dn fake.fake -enc binary
```

The following example shows the `enroll_resume` command.

```
cisco_cert_mgr -E -op enroll_resume -ct 4
```

Table 5-2 describes keywords for the `enroll`, `enroll_file`, and `enroll_resume` operations.

Table 5-2 Keywords for Enrollment Operations

Keyword	Description
-cn <i>common_name</i>	The common name for the certificate.
-ou <i>organizational_unit</i>	The organizational unit for the certificate.
-o <i>organization</i>	The organization for the certificate.
-st <i>state</i>	The state for the certificate.
-c <i>country</i>	The country for the certificate.
-e <i>email</i>	The user e-mail address for the certificate.
-ip <i>IP_Address</i>	The IP address of the user's system.
-dn <i>domain_name</i>	The fully qualified domain name of the user's system.
-caurl <i>url_of_CA</i>	The URL or network address of the CA.
-cadn <i>domain_name</i>	The CA's domain name.

Table 5-2 Keywords for Enrollment Operations (continued)

Keyword	Description
<code>[-chall challenge_phrase]</code>	You can obtain the challenge phrase from your administrator or from the CA.
<code>-enc [base64 binary]</code>	Select encoding of the output file. The default is base64. • base64 is an ASCII-encoded PKCS10 file that you can display because it is in a text format. Choose this type when you want to cut and paste the text into the CA's website. • binary is a base-2 PKCS10 (Public-Key Cryptography Standards) file. You cannot display a binary-encoded file.

Enrollment Troubleshooting Tip

If the enrollment request for a user certificate, using either the `enroll` or `enroll_file` operation, generates a CA certificate instead of a user certificate, the CA might be overwriting some of the distinguished naming information. This might be caused by a configuration issue on the CA, or a limitation of how the CA responds to enrollment requests.

The common name and subject information in the enrollment request must match the certificate generated by the CA for the VPN client to recognize it as the same user certificate that was requested. If it does not match, the VPN client does not install the new user certificate as the user certificate it had requested.

To check for this problem, view the enrollment request on the VPN client and compare the common name and subject lines with a view of the certificate from the CA. If they do not match, then the CA is overwriting information from the client request.

To work around this issue, use the invalid certificate as an example and create an enrollment request that matches the output of the CA certificate.

**Note**

If the CA's certificate contains multiple department (multiple ou fields), you can add multiple departments to the VPN client enrollment request by using the plus sign (+) between the department fields.



Preconfiguring the VPN Client for Remote Users

This chapter explains how to create and edit global profiles.

A series of configuration parameters determine the user profiles that remote users choose to connect to a VPN device. These profiles have a .pcf file extension and the default location is /etc/CiscoSystemsVPNClient/Profiles.

There is also a global profile that you can use to set certain standards for all user profiles. The name of the global profile file is vpnclient.ini. You can create a global profile that contains preconfigured information for a group of users.

For information on user profiles, see Chapter 3, “User Profiles.”

Contents

This chapter contains the following sections:

- Making a Parameter Read-only, page 6-2
- Creating a Global Profile, page 6-2
- Limiting User Access, page 6-4
- Distributing Preconfigured VPN Client Software, page 6-5

Making a Parameter Read-only

To set a parameter to be read-only so that a user cannot change it within the VPN client applications, precede the parameter name with an exclamation mark (!). This only controls the action the user can take *within* the VPN client applications. You cannot prevent someone from editing the global file or removing it.

Creating a Global Profile

The global profile, `vpnclient.ini`, resides in the `/etc/CiscoSystemsVPNClient/` directory. This is the default location and is created during installation.

The following is an example of a global profile opened using a text editor.

```
[main]
RunAtLogon=0
EnableLog=1
[LOG.IKE]
LogLevel=1
[LOG.CM]
LogLevel=1
[LOG.PPP]
LogLevel=2
[LOG.DIALER]
LogLevel=2
[LOG.CVPND]
LogLevel=1
[LOG.CERT]
LogLevel=0
[LOG.IPSEC]
LogLevel=3
[CertEnrollment]
SubjectName=Alice Wonderland
Company=University of OZ
Department=International Relations
State=Massachusetts
Country=US
Email=AliceW@UOZ.com
CADomainName=CertsAreUs
CAHostAddress=10.10.10.10.
CACertificate=CAU
```

Global Profile Configuration Parameters

This section describes the parameters that can be configured in the global profile. Table 6-1 lists all global profile keywords and their descriptions.

Table 6-1 Global Profile Keywords

Keyword	Description
[main]	This keyword is required and is used to identify the main section. Enter exactly as shown as the first entry in the user profile.
EnableLog = { 0 1 }	Determines whether or not to override log settings for the classes that use logging services. By default, logging is turned on. Use this parameter if you want to disable logging without having to set the log levels to zero for each of the classes. • 0 disables logging services. • 1 enables logging services. You can improve the performance of the VPN client system by turning off logging.
BinDirPath = <i>String</i>	The path to where the VPN client was installed. The maximum value is 512 characters. The default is /usr/local/bin.
LogLevel = { 0 1 2 3 }	Determines the log level for individual classes that use logging services. • 0 disables logging services for the specified [LOG] class. • 1, low, displays only critical and warning events. This is the default. • 2, medium, displays critical, warning, and informational events. • 3, high, displays all events.

Table 6-1 Global Profile Keywords (continued)

Keyword	Description
Use the LogLevel parameter to set the logging level for each of the following LOG classes. Enter the parameter as shown.	
[LOG.IKE]	Identifies the IKE class for setting the logging level.
[LOG.CM]	Identifies the CM class for setting the logging level.
[LOG.CVPND]	Identifies the CVNPD class for setting the logging level.
[LOG.CERT]	Identifies the CERT class for setting the logging level.
[LOG.IPSEC]	Identifies the IPsec class for setting the logging level.
CACertificate = <i>String</i>	Identifies the name of the self-signed certificate issued by the certificate authority (CA). The maximum length is 519 alphanumeric characters.
NetworkProxy = <i>IP_Address or</i> <i>hostname</i>	This keyword is used for SCEP (certificate enrollment). Identifies a proxy server you can use to route HTTP traffic. Using a network proxy can help prevent intrusions on your private network. The maximum length of the hostname is 519 alphanumeric characters. The proxy setting might have a port associated with it. If so, enter the port number after the IP address. For example, 10.10.10.10.8080.

Limiting User Access

Upon installation, any user on your system can establish a VPN connection, or view, edit, and add user profiles. You can limit users from accessing certain files, and limit their ability to establish a connection.

To limit access to the VPN client to only the root user, issue the following commands:

```
% chmod 700 /usr/local/bin/vpnclient
% chmod 700 /usr/local/bin/cvpnd
% chmod 700 /etc/CiscoSystemsVPNClient/Profiles
% chmod 700 /etc/CiscoSystemsVPNClient/Certificates
```

```
% chmod 700 /etc/CiscoSystemsVPNClient/vpnclient.ini
% chmod 700 /etc/CiscoSystemsVPNClient/Profiles/*
```

To limit access to profile information but allow a VPN connection, issue the following commands:

```
% chmod 700 /etc/CiscoSystemsVPNClient/Profiles
% chmod 700 /etc/CiscoSystemsVPNClient/Certificates
% chmod 700 /etc/CiscoSystemsVPNClient/vpnclient.ini
% chmod 700 /etc/CiscoSystemsVPNClient/Profiles/*
```

To reset the VPN client and return the `cvpnd`, `vpnclient.ini`, profiles, and certificate directories back to default permissions, rerun the VPN install script. For more information on running the install script, see the “Installing the Software” section on page 2-6.

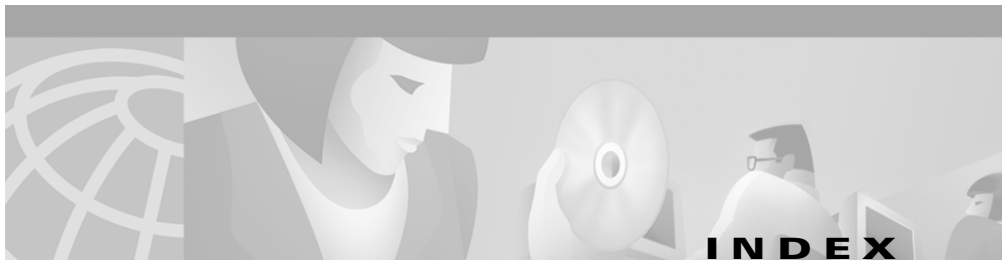
Distributing Preconfigured VPN Client Software

This section describes how to distribute the preconfigured VPN client profiles to users along with the installation software. You can distribute the VPN client global profile or user profile to users separately or as part of the VPN client software.

Separate Distribution

To distribute the profiles separately and have users import them into the VPN client after they have installed it on their PCs:

- a. Distribute the appropriate profile files to users on whatever media you prefer.
- b. Supply users with necessary configuration information.
- c. Instruct users to:
 - Install the VPN client according to the instructions in Chapter 2, “Installing the VPN Client.”
 - Modify their user profile as described in Chapter 3, “User Profiles.”
 - Establish a VPN client connection as described in Chapter 4, “Using the Command Line Interface.”



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